



Urban green spaces and air quality in the State of São Paulo: a spatiotemporal analysis of the metropolitan regions of Campinas and Piracicaba

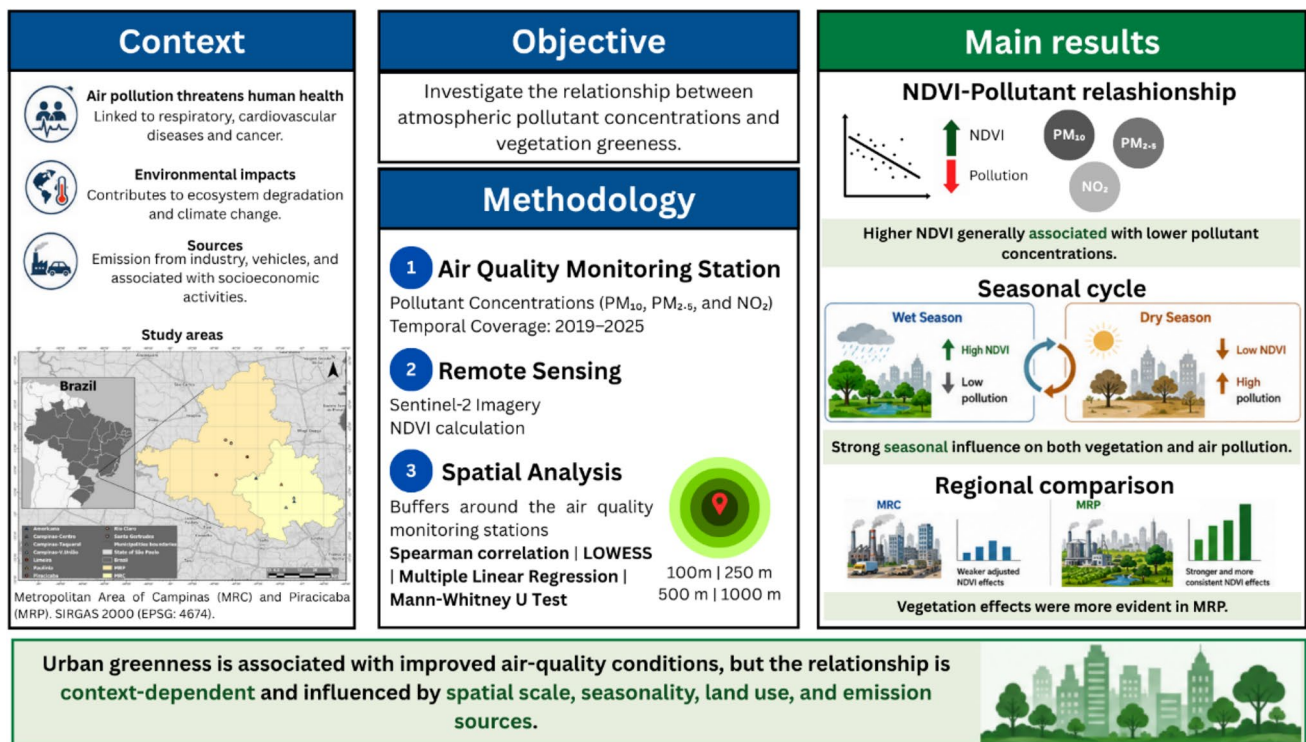
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Abstract

Air pollution remains a major environmental and public-health concern, especially in urban and peri-urban regions where emissions, land use, and seasonal meteorology interact. Although GI is often proposed as a strategy to improve air quality, its effects are highly context-dependent and difficult to isolate. This study examined the association between vegetation greenness, represented by NDVI, and monthly concentrations of particulate matter (PM₁₀ and PM_{2.5}), and NO₂ in the MRC and MRP, São Paulo State, Brazil, from 2019 to 2025. Pollutant and meteorological data were obtained from CETESB AQMS, and NDVI was derived from Sentinel-2 imagery in Google Earth Engine within 100, 250, 500, and 1000 m buffers around each AQMS. Associations were assessed using Spearman correlations, LOWESS smoothing, Mann–Whitney U tests, and multiple linear regression models adjusted for meteorological variables and temporal fixed effects. NDVI was predominantly negatively correlated with pollutant concentrations, with coefficients ranging from -0.24 to -0.83 for PM₁₀, -0.17 to -0.82 for PM_{2.5}, and -0.31 to -0.75 for NO₂. Seasonal patterns were evident, with higher pollutant concentrations and lower NDVI values during the dry season. In adjusted regression models, significant negative NDVI associations were mainly observed in MRP, particularly for PM₁₀ at 100–250 m, PM_{2.5} at 100–500 m, and NO₂ at 100–500 m, whereas most associations in MRC were non-significant. PM₁₀ and NO₂ concentrations were significantly higher in MRP than in MRC, whereas PM_{2.5} showed no significant regional difference. These findings suggest that urban greenness is associated with air quality patterns in a context-dependent manner, reinforcing the need to account for spatial scale, seasonality, and local land use conditions in GI planning.

Graphical abstract



Keywords Green infrastructure · Particulate matter · Nitrogen dioxide · Google earth engine

Abbreviations

AIC	Akaike information criterion
AOI	Area of Interest
API	Application Programming Interface
AQMS	Air Quality Monitoring Station(s)
CETESB	São Paulo State Environmental Agency
GEE	Google Earth Engine
GI	Green Infrastructure
HAC	Heteroskedasticity and Autocorrelation Consistent
IBGE	Brazilian Institute of Geography and Statistics
LOWESS	Locally Weighted Scatterplot Smoothing
LULC	Land Use and Land Cover
MLR	Multiple Linear Regression
MRC	Metropolitan Region of Campinas
MRP	Metropolitan Region of Piracicaba
MSI	Multispectral Instrument
NDVI	Normalized Difference Vegetation Index
NO ₂	Nitrogen Dioxide
PM	Particulate Matter

PM ₁₀	Particulate Matter with Aerodynamic Diameter less than 10 µm
PM _{2.5}	Particulate Matter with Aerodynamic Diameter less than 2.5 µm
RH	Relative Humidity
TEMP	Air Temperature
UHI	Urban Heat Island
VIF	Variance Inflation Factor
WHO	World Health Organization
WS	Wind Speed

Introduction

Air pollution is a global challenge with direct impacts on public health, being associated with an increased risk of respiratory and cardiovascular diseases, as well as cancer. Furthermore, it contributes to ecosystem degradation and climate change through the emission and dispersion of pollutants across environmental compartments (Manisalidis et al. 2020).

Its composition is generally characterized by the presence of particulate matter (PM), gases, and vapors originating from industrial and vehicular emissions, as well as broader socioeconomic activities and consumption patterns (World Health Organization 2021).

In urban areas, air quality is further influenced by the configuration of buildings, urban density, and the distribution of transportation networks, all of which affect patterns of ventilation, pollutant dispersion, and atmospheric pollutant accumulation (Srbinovska et al. 2025). Understanding these relationships is important for developing effective mitigation strategies (Jo et al. 2025).

One alternative for mitigating atmospheric pollution is the implementation of natural or semi-natural systems based on vegetation cover, such as green infrastructure (GI) (Benedict et al. 2006; Tzoulas et al. 2007). The GI can contribute to the removal of atmospheric pollutants through regulating functions, while also reducing air temperature and, consequently, mitigating the effects of urban heat islands (UHI) (Srbinovska et al. 2025).

These functions operate through specific biophysical mechanisms. PM is primarily removed through dry deposition on leaf surfaces, where canopy structure, leaf area, and surface roughness determine particle interception and retention efficiency, while gaseous pollutants such as NO_2 are taken up mainly through stomatal absorption during gas exchange (Janhäll 2015). However, these same mechanisms can be offset by aerodynamic effects, when dense canopies in confined urban settings may reduce wind flow and pollutant dispersion, occasionally leading to localized pollutant accumulation rather than removal (Janhäll 2015). This duality highlights that the net effect of vegetation on air quality depends not only on vegetation abundance but also on canopy structure and urban morphology.

Some studies have reinforced the association between vegetation presence and improved air quality. Srbinovska et al. (2025) reported that areas with greater vegetation cover exhibited reductions of up to 37% in particulate matter concentrations. Similarly, Matthaios et al. (2024) analyzed data from 74 urban schools and found that schools located farther

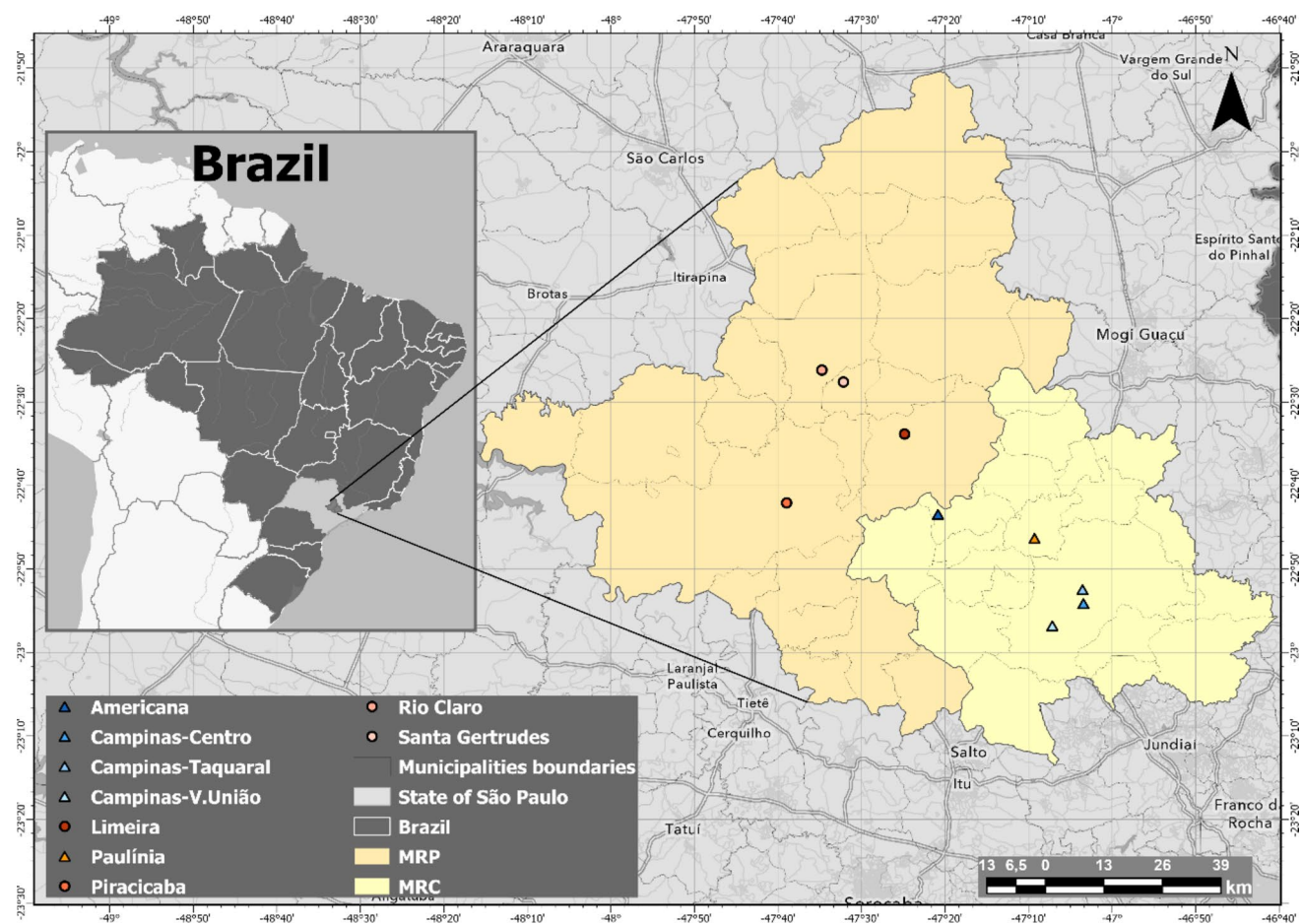


Fig. 1 Study areas in the MRC and MRP, São Paulo State, Brazil. The map highlights municipal boundaries, the delimitation of the metropolitan regions, and the study areas in relation to the national

territory. Reference System: SIRGAS 2000 (EPSG: 4674); Coordinate System: geographic coordinates (latitude and longitude in decimal degrees). Data source: based on (IBGE 2022)

from high-traffic roads and surrounded by green spaces exhibited lower concentrations of atmospheric pollutants in indoor environments.

Despite these documented benefits, the magnitude of vegetation effects on air quality remains highly dependent on local environmental and spatial characteristics, making it difficult to generalize findings across urban contexts (Balzan et al. 2018; Preston et al. 2024).

The strength of these associations is also expected to vary with the spatial scale considered, since the influence of surrounding vegetation on local pollutant concentrations may differ substantially depending on the distance from emission sources and monitoring points. Previous studies modeling the effects of vegetation and urban form on air quality have shown that this relationship is scale-dependent, with vegetation effects sometimes only becoming detectable beyond certain spatial thresholds (Matthaios et al. 2024; Yao et al. 2025). This reinforces the need for multi-scale analytical approaches capable of identifying the spatial scales at which vegetation–pollutant associations are detectable.

Remote sensing technologies have substantially improved the capacity to monitor vegetation dynamics and land use and land cover (LULC) patterns across large spatial and temporal scales (Schaefer et al. 2021; Shi et al. 2024). Because LULC influences the spatial distribution of emission sources, pollutant dispersion processes, and the availability of green spaces, it plays an important role in shaping atmospheric pollutant concentration patterns (Tao et al. 2018; Brauer et al. 2019; Sannigrahi et al. 2020; Jodhani et al. 2024). These advances provide valuable opportunities to investigate vegetation–air quality relationships using spatially explicit indicators derived from remotely sensed data.

Vegetation cover in this context is commonly assessed using the Normalized Difference Vegetation Index (NDVI),

a remote sensing-based indicator derived from the spectral reflectance of vegetation that reflects canopy density (Gao et al. 2020; Huang et al. 2021). It is important to note that NDVI is an indirect structural proxy for vegetation vigor, rather than a direct measurement of pollutant filtration or air purification capacity.

In this regard, the central hypothesis of this study is that greater vegetation cover is associated, to some extent, with lower concentrations of atmospheric pollutants. Based on this premise, this study aims to investigate the relationship between atmospheric pollutant concentrations and vegetation cover, represented by NDVI, across two metropolitan regions in the state of São Paulo, Brazil, over a seven-year period (2019–2025).

Methods

Study area

This study focuses on two adjacent metropolitan regions in the State of São Paulo, southeastern Brazil: Metropolitan Region of Campinas (MRC) and Metropolitan Region of Piracicaba (MRP) (Fig. 1). These regions were selected due to their strategic economic roles, contrasting LULC patterns, which together can provide a framework for assessing the effects of urbanization and GI on air quality.

The MRC, comprising 20 municipalities, is characterized by a high degree of urbanization and intense industrial, technological, logistics, and service-sector activities, resulting in substantial vehicular and industrial emissions. Its landscape reflects this profile, with urban, agricultural, and mixed-use areas occupying comparable proportions of the territory (Souza et al. 2020). In contrast, the MRP,

Table 1 AQMS included in the study, monitored pollutants, and surrounding LULC context (1000 m buffer)

Region	AQMS	UTM Coordinates	CETESB SR	UD	R	C	I	T	GS
MRC	Americana	23 K 258896 7485068	NS (a)	4	4	2	1	3	3
	Campinas-Centro	23 K 288991 7465796	MiS (a)	4	1	4	0	4	1
	Campinas-Taquaral	23 K 288767 7468884	NS (a, c)	4	4	2	0	2	4
	Campinas-V.União	23 K 282693 7460810	US (b)	4	4	1	2	4	1
	Paulínia	23 K 278763 7480073	NS (a, c)	3	4	2	1	3	2
MRP	Limeira	23 K 251737 7502780	NS (a, b, c)	4	4	2	1	3	2
	Piracicaba	23 K 227797 7487124	NS (a, b, c)	3	2	1	4	4	3
	Rio Claro	23 K 234497 7516625	NS (a, b)	4	4	2	2	4	2
	Santa Gertrudes	23 K 238993 7514053	NS (a, b, c)	3	4	2	3	4	3

(a) PM₁₀; (b) PM_{2.5}; (c) NO₂. CETESB SR, CETESB spatial representativeness classification; MiS, micro-scale; NS, neighborhood-scale; US, urban-scale. Land use classes are abbreviated as follows: UD, urban density; R, residential; C, commercial; I, industrial; T, traffic; and GS, green space. Coordinates are provided in the UTM projection (Zone 23 K), referenced to the SIRGAS 2000 datum. All AQMS are automatic operated by CETESB. Source: CETESB (2024). Land use intensity was classified using a five-level scale: 0 = absent; 1 = very low; 2 = low; 3 = moderate; 4 = high; and 5 = very high (or dominant). Sources: (CETESB 2016, 2024a, b)

consisting of 24 municipalities, is predominantly characterized by agro-industrial activities, particularly those associated with sugarcane cultivation and processing, with crop-land and mixed agricultural uses representing the dominant land cover classes (Souza et al. 2020).

Both regions are located within the Atlantic Forest biome, one of the world's biodiversity hotspots (Myers et al. 2000; Souza et al. 2020), which has undergone extensive fragmentation as a result of urban expansion, agricultural development, and industrial infrastructure growth. Consequently, remaining forest fragments and urban green spaces are important in microclimate regulation, particularly in densely populated areas. The regional climate is classified as humid subtropical, with hot, rainy summers and dry winters (Alvares et al. 2013), conditions that strongly influence pollutant dispersion, secondary pollutant formation, and vegetation dynamics throughout the year (Seinfeld and Pandis 2006).

The analyzed pollutants included particulate matter with an aerodynamic diameter of less than 10 μm (PM_{10}), particulate matter with an aerodynamic diameter of less than 2.5 μm ($\text{PM}_{2.5}$), and nitrogen dioxide (NO_2). All data were obtained from CETESB, which operates an automated monitoring network composed of fixed stations spread across the São Paulo Metropolitan Region, inland areas, and coastal zones, as well as mobile units for temporary assessments. The network continuously measures regulated pollutant concentrations through telemetric systems that transmit real-time data to a central database, with hourly updates following standardized protocols in compliance with Brazilian air quality regulations.

Given that pollutant availability varied across stations, Table 1 summarizes the air quality monitoring stations (AQMS) included in the study, their geographic locations, the pollutants monitored at each site, and the surrounding LULC characteristics within a 1000 m buffer, which were classified through visual interpretation of satellite imagery.

Furthermore, CETESB classifies AQMS according to pollutant-specific spatial representativeness, including micro-scale, middle-scale, neighborhood-scale, and urban-scale categories (CETESB 2016). Therefore, the spatial representativeness classification of each AQMS used in this study was verified and also reported in Table 1.

Air quality monitoring data

Data were obtained from CETESB's QUALAR platform, using automated AQMS, an open-access dataset that provides validated air pollution measurements from the official monitoring network in São Paulo State, Brazil.

Hourly atmospheric pollutant concentration data and meteorological parameters were obtained for a seven-year period (2019–2025). The data were processed at their original hourly resolution (24 measurements per day at each

monitoring station). Initially, data organization and quality control procedures were applied to ensure consistency and suitability for subsequent analyses. Negative values and concentrations exceeding predefined physical thresholds (1000.00 $\mu\text{g m}^{-3}$ for PM_{10} , 500.00 $\mu\text{g m}^{-3}$ for $\text{PM}_{2.5}$, and 400.00 $\mu\text{g m}^{-3}$ for NO_2) were removed, as these values were considered indicative of spurious measurements resulting from instrumental malfunctions or data transmission errors.

To address data completeness and reduce the influence of missing observations, daily averages were computed from the hourly data. A minimum data availability criterion was applied, whereby a daily mean was considered valid only if at least 75% of the expected hourly measurements (i.e., at least 18 out of 24 h) were available for that day. Days that did not meet this requirement were flagged as invalid, and their daily mean concentrations were set to missing values. This approach ensures that daily statistics are representative of actual atmospheric conditions and minimizes bias due to incomplete temporal coverage.

Subsequently, daily data were aggregated into monthly statistics for each monitoring station and pollutant, calculated based on the valid daily values. To ensure temporal representativeness at the monthly scale, a completeness criterion was also applied: months were considered valid only if at least 75% of the expected number of days contained valid daily data. Further details on data availability for each AQMS are provided in the Supplementary Material (Fig. SM1).

Vegetation cover patterns

To characterize vegetation cover patterns surrounding the AQMS, the Normalized Difference Vegetation Index (NDVI) was used as a proxy for vegetation density and condition (Dadvand et al. 2015; Matthaïos et al. 2024). Monthly mean NDVI values were calculated within concentric buffer zones of 100 m, 250 m, 500 m, and 1000 m around each monitoring station.

The selection of these buffer distances aimed to represent different spatial scales at which vegetation may influence atmospheric pollutant concentrations. This multi-buffer approach has been widely adopted in studies investigating the relationships between urban greenness, environmental exposure, and air quality, as the strength and direction of these associations may vary according to the spatial scale considered (Dadvand et al. 2015; Jimenez et al. 2022; Matthaïos et al. 2024). The 100 m buffer was used to characterize vegetation immediately adjacent to the monitoring stations, potentially associated with local effects such as particle interception and modifications in pollutant dispersion patterns. The 250 m buffer represents the station's nearby surroundings. The 500 m and 1000 m buffers were used to capture neighborhood-scale urban landscape characteristics,

including green space connectivity, urban density, and traffic intensity.

Thus, satellite imagery was obtained from the Harmonized Sentinel-2 Multispectral Instrument (MSI) collection, which provides surface reflectance data with a spatial resolution of 10 m. The analysis covered the same period considered for the atmospheric pollutant dataset.

Sentinel-2 imagery provides 10 m spatial resolution and an approximate 5-day revisit time. In this study, cloud-masked scenes were aggregated into monthly mean NDVI values to match the temporal resolution of the pollutant dataset. Although this resolution is suitable for estimating NDVI within 100 m buffers, it may underrepresent small or fragmented urban vegetation due to mixed-pixel effects.

The image collection was temporally filtered to include scenes acquired between January and December of each analyzed year and spatially clipped to the predefined areas of interest (AOI) corresponding to the buffer zones around the monitoring stations. To reduce atmospheric and cloud-related influence, cloud and cirrus masking algorithms were applied, and only images with less than 15% cloud cover were retained. Optical bands (blue, green, red, and near-infrared) were rescaled using the provided scale factors to convert reflectance values to a standardized 0–1 range.

Vegetation greenness was inferred from the spectral contrast between red and near-infrared reflectance, since healthy vegetation absorbs red radiation and strongly reflects near-infrared radiation. NDVI was calculated as the normalized difference between the near-infrared band (Sentinel-2 Band 8) and the red band (Sentinel-2 Band 4). Monthly mean NDVI values were then derived for each AOI. All image processing and subsequent analyses were performed in Google Colab using the Google Earth Engine (GEE) API.

Data analysis

The data described in “Air quality monitoring data” and “Vegetation cover patterns” sections were merged into a single dataset. Prior to statistical analyses, data distributions were evaluated using descriptive statistics and visual inspection to assess their characteristics and suitability for subsequent statistical analyses. Because exploratory diagnostics indicated skewed distributions, nonparametric methods were used in environmental datasets.

Spearman correlation analysis

Pairwise associations between monthly pollutant concentrations and NDVI were assessed using Spearman’s correlation coefficient (ρ) (Hollander et al. 2015), which is suitable for

data that do not follow a normal distribution, as is the case for most environmental data, and is robust to the presence of outliers, a common characteristic of atmospheric data series (Wilks 2006). Correlations were calculated separately for each monitoring station and each buffer radius, producing a matrix of ρ values that captures the spatial heterogeneity of the relationship between vegetation and air quality. Statistical significance was assessed at three levels ($p < 0.05$; $p < 0.01$; and $p < 0.001$), with a minimum of 10 valid monthly pairs per station required to compute each coefficient. All tests were two-tailed.

Multiple linear regression

To quantify the adjusted association between NDVI and pollutant concentrations, after controlling for meteorological variability, multiple linear regression (MLR) models were fitted for each pollutant–region–buffer combination. Given the typically log-normal distribution of atmospheric pollutant concentrations (Seinfeld and Pandis 2006), the natural logarithm was used as the response variable. All pollutants were log-transformed using this same approach. Prior to transformation, the monthly dataset was inspected for zero or near-zero values, and none were found for any pollutant in either metropolitan region, and the natural logarithm was therefore applied directly, without requiring an offset constant. Fixed effects for month and year were included as categorical predictors to account for cyclical seasonal patterns and long-term inter-annual trends, respectively. The general model specification is presented in Eq. 1.

$$\ln(C_{i,t}) = \beta_0 + \beta_1 NDVI_{i,t} + \beta_2 TEMP_{i,t} + \beta_3 RH_{i,t} + \beta_4 WS_{i,t} + \beta_5 Month_t + \beta_6 Year_t + \epsilon_{i,t} \quad (1)$$

where C is the mean monthly concentration at monitoring station i in month t ; NDVI is the mean vegetation index within the buffer of interest; TEMP, RH, and WS correspond, respectively, to mean air temperature ($^{\circ}\text{C}$), mean relative humidity (%), and mean wind speed (m s^{-1}); Month and Year are categorical fixed effects; and ϵ is the error term. Prior to model fitting, multicollinearity among predictors was assessed using the Variance Inflation Factor (VIF) and predictors with $VIF > 10$ were excluded.

To assess scale dependency, separate regression models were fitted independently for each buffer radius (100, 250, 500, and 1000 m). In each model, NDVI was included as a predictor using one fixed buffer radius at a time, while buffer size itself was not included as a covariate or interaction term. The unit of analysis was the monitoring station month observation (i, t). This approach avoided including multiple highly correlated NDVI metrics from overlapping,

Nonlinear relationship between pollutants concentration and NDVI

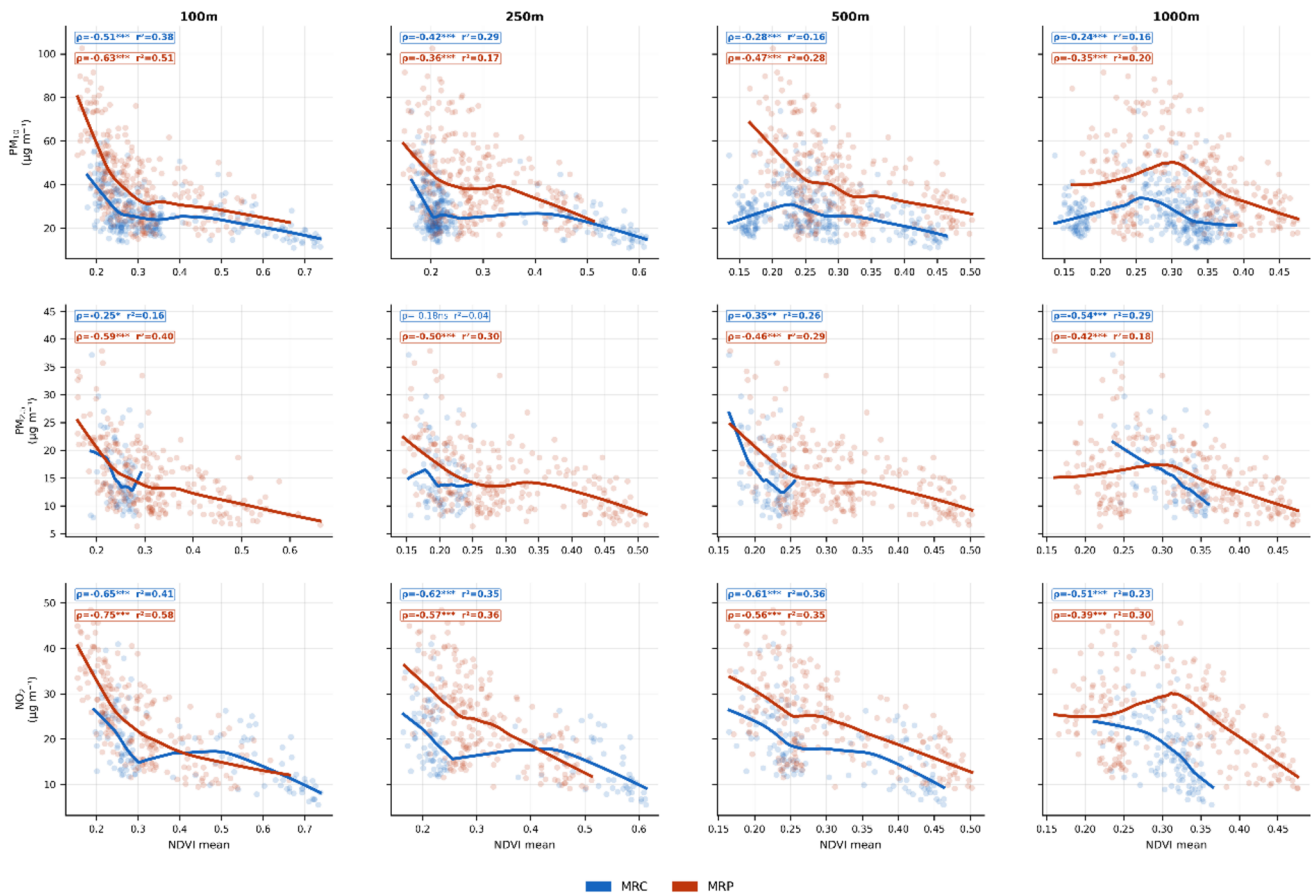


Fig. 2 Relationships between pollutant concentrations and NDVI across buffer distances of 100, 250, 500, and 1000 m for the MRC and the MRP. Smoothed curves were fitted using locally weighted scatterplot smoothing (LOWESS). Spearman's rank correlation coef-

ficient (ρ) and pseudo- r^2 are reported for each region and buffer zone. Asterisks indicate statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

nested buffer geometries in the same model, thereby reducing multicollinearity concerns.

To formally address potential serial autocorrelation arising from the monthly, multi-year structure of the dataset, model inference was based on heteroscedasticity- and autocorrelation-consistent (HAC) standard errors, following the Newey-West estimator (Newey and West 1987), with a maximum lag of one month (maxlag = 1).

Model performance was evaluated using the adjusted coefficient of determination (adjusted R^2) and the Akaike Information Criterion (AIC), which were compared across the four buffer-specific models to assess the sensitivity of the results to the spatial scale of the vegetation metric.

Comparison between regions

Differences in pollutant concentration distributions between the MRC and MRP were assessed using the two-sided

Mann–Whitney U test, a nonparametric alternative to the independent-samples t test that does not assume normality. The null hypothesis of equal distributions was tested separately for each pollutant, adopting a significance level of $\alpha = 0.05$. This analysis complements the within-station correlation and regression results by quantifying whether the two metropolitan regions present distinct air quality conditions, which may act as a moderator of the magnitude of the vegetation–air pollution relationship.

Results

The relationships between monthly mean NDVI and atmospheric pollutant concentrations (PM₁₀, PM_{2.5}, and NO₂) across all buffers (100, 250, 500, and 1000 m) for the MRC and MRP, fitted with LOWESS smoothing curves, consistently present negative correlations (Fig. 2). The smoothed

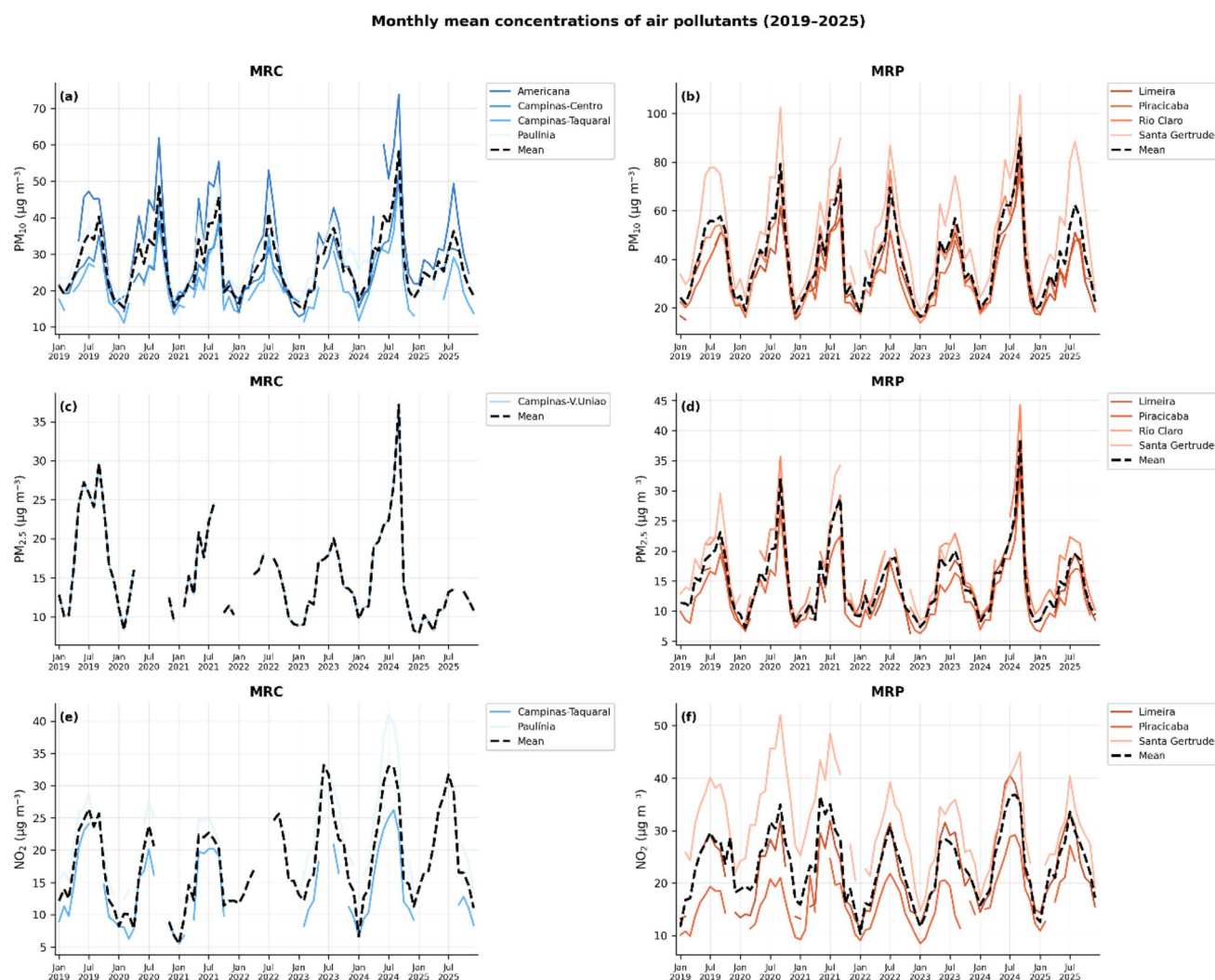


Fig. 3 Monthly mean concentrations of PM₁₀ (a, b), PM_{2.5} (c, d), and NO₂ (e, f) measured at AQMS (colored lines) and regional means (dashed black lines) in the MRC and MRP from January 2019 to December 2025

curves consistently deviated from linearity, displaying a steep decline at low NDVI values (<0.30) that progressively flattened with increasing vegetation density, a pattern observed across both regions.

PM₁₀ presented significant negative correlations in both regions, with ρ ranging from -0.24 to -0.51 in MRC and from -0.35 to -0.63 in MRP ($p < 0.001$), and the curves also displaying a concave shape with steeper gradients below NDVI = 0.30. For PM_{2.5}, associations in MRC were weaker and buffer-dependent, ranging from non-significant at 250 m ($\rho = -0.18$, $p = 0.14$) to moderate at 1000 m ($\rho = -0.54$, $p < 0.001$), whereas in MRP correlations were consistently significant across all buffers ($\rho = -0.42$ to -0.59 , $p < 0.001$). NO₂ showed the strongest associations with NDVI, with Spearman's ρ ranging from -0.51 to -0.65 in MRC and from -0.39 to -0.75 in MRP ($p < 0.001$).

Temporal dynamics

Monthly mean concentrations of pollutants exhibited a consistent seasonal cycle throughout the study period (2019–2025), with higher concentrations during the dry season, particularly between May and August, and lower concentrations during the wet season, mainly from December to February (Fig. 3). This pattern reflects the reduced precipitation and lower wind speeds of the dry season, which suppress wet deposition and limit pollutant dispersion.

Dry-to-wet season ratios revealed that PM₁₀ experienced the strongest seasonal contrast in MRP, where dry season median concentrations ($53.59 \mu\text{g m}^{-3}$) were 2.5 times higher than wet season values ($21.65 \mu\text{g m}^{-3}$). In MRC, the ratio was 1.8 times (32.96 vs. $18.32 \mu\text{g m}^{-3}$). For PM_{2.5}, seasonal amplitudes were more comparable between regions (~ 1.9 times in MRC and ~ 2.0 times in MRP). NO₂ showed

Seasonal patterns of air pollutants and NDVI (2019–2025)

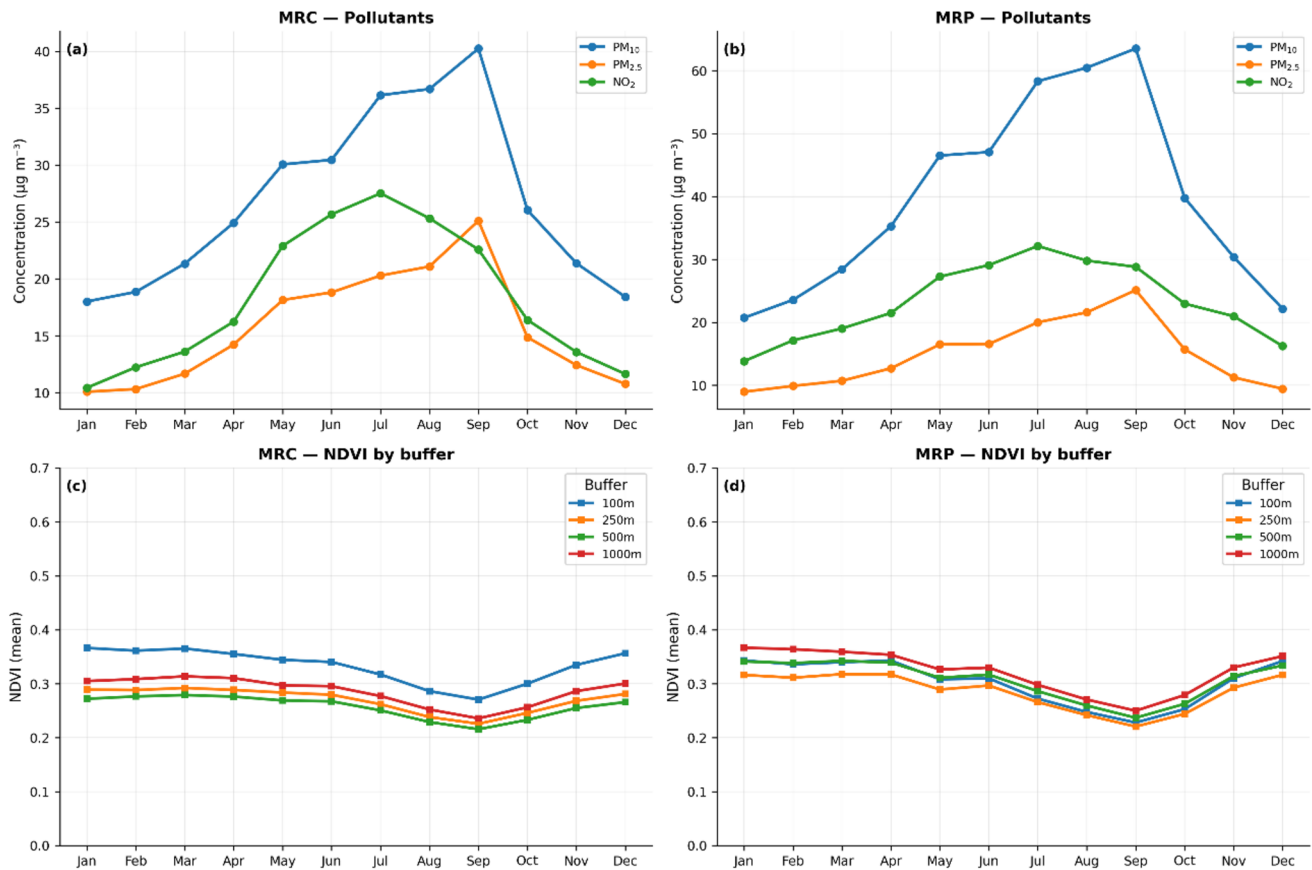


Fig. 4 Seasonal monthly patterns of air pollutants and NDVI at AQMS (2019–2025). Upper panels (a, b) show mean monthly concentrations of particulate matter (PM_{10} and $PM_{2.5}$, in $\mu g m^{-3}$) and

nitrogen dioxide (NO_2 , in $\mu g m^{-3}$). Lower panels (c, d) display mean monthly NDVI values calculated within buffer zones of 100, 250, 500, and 1000 m around each AQMS

Spearman's ρ between pollutants and NDVI by monitoring station and buffer
(* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$ | dashed line separates MRC from MRP)

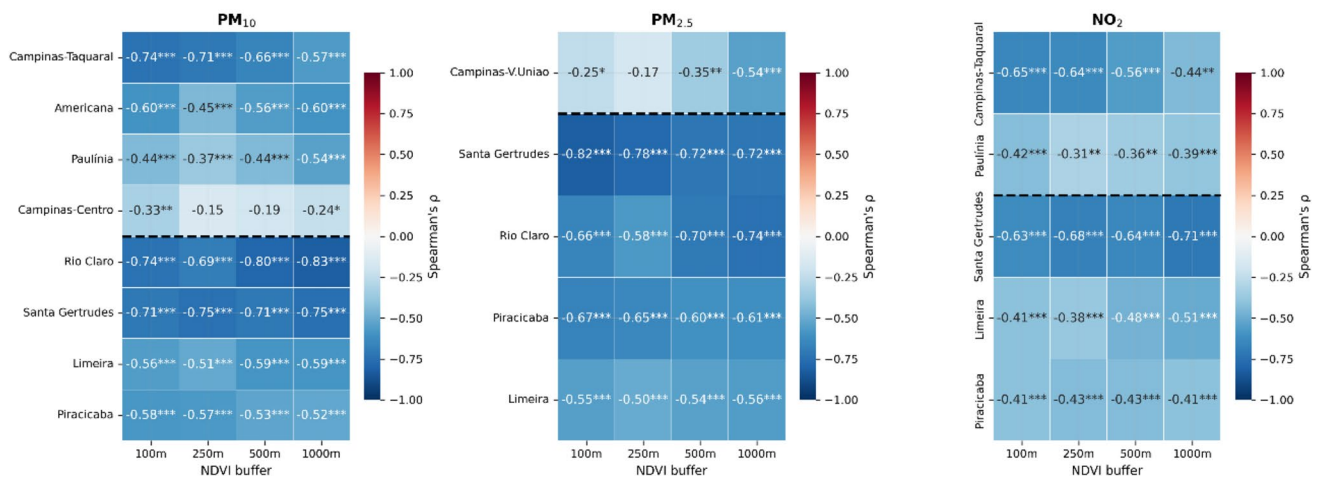


Fig. 5 Spearman's rank correlation coefficients (ρ) between NDVI and air pollutant concentrations across AQMS and NDVI buffer sizes (100, 250, 500, and 1000 m). Color intensity reflects the magnitude of the correlation, ranging from weak (light blue) to strong (dark

blue) negative associations. Asterisks indicate statistical significance (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). The dashed horizontal line separates MRC (above) from MRP (below) monitoring stations

amplitudes of approximately 2.0 times in MRC and 1.8 times in MRP.

Monthly medians indicated that peak concentrations for PM_{10} and $PM_{2.5}$ occurred in September in both regions, exceeding dry season medians despite the return of rainfall. In MRC, September medians reached $40.31 \mu\text{g m}^{-3}$ for PM_{10} and $23.59 \mu\text{g m}^{-3}$ for $PM_{2.5}$, while in MRP they reached 57.67 and $23.14 \mu\text{g m}^{-3}$, respectively. This late-season peak likely reflects the influence of long-range transport of biomass burning aerosols from central Brazil, whose fire activity typically intensifies through August and September.

NO_2 peaked earlier, in July (MRC: $29.10 \mu\text{g m}^{-3}$; MRP: $31.58 \mu\text{g m}^{-3}$), and declined progressively through September, reflecting its stronger dependence on atmospheric mixing conditions rather than fire emissions. Minimum concentrations for all three pollutants were consistently recorded in January, at the height of the wet season (MRC: $PM_{10} = 17.48$, $PM_{2.5} = 9.73$, $NO_2 = 11.76 \mu\text{g m}^{-3}$; MRP: $PM_{10} = 21.05$, $PM_{2.5} = 9.20$, $NO_2 = 12.59 \mu\text{g m}^{-3}$).

Despite the dominant seasonal signal, notable inter-annual variability was observed. Based on annual medians of the regional monthly mean concentrations, PM_{10} concentrations in MRC ranged from $22.34 \mu\text{g m}^{-3}$ (2021) to $30.08 \mu\text{g m}^{-3}$ (2024), while in MRP they ranged from $32.46 \mu\text{g m}^{-3}$ (2021) to $42.66 \mu\text{g m}^{-3}$ (2024), with 2024 standing out as an anomalously polluted year in both regions, likely associated with the exceptional drought and intensified fire activity recorded across Brazil that

year. In contrast, NO_2 inter-annual variability was more pronounced in MRC, where annual medians dropped to $10.09 \mu\text{g m}^{-3}$ in 2020, likely reflecting reduced traffic and industrial activity during the COVID-19 pandemic, before increasing again and reaching $18.65 \mu\text{g m}^{-3}$ in 2023. In MRP, NO_2 levels remained comparatively stable across years (20.45 – $25.15 \mu\text{g m}^{-3}$).

The seasonal dynamics of NDVI mirrored, in inverse fashion, the pollutant concentration cycles (Fig. 4). Across all buffer zones, NDVI peaked during January–March and reached its minimum in September, precisely when PM_{10} and $PM_{2.5}$ concentrations were at their highest. This temporal anti-phase between vegetation greenness and pollutant accumulation was consistent across both regions.

Relationship between NDVI and Pollutants

Spearman's correlation analysis indicated predominantly negative associations between NDVI and pollutant concentrations, although the magnitude and statistical significance varied among stations, pollutants, and buffer sizes (Fig. 5).

For PM_{10} , significant negative correlations ($p < 0.001$) were observed at the majority of stations. The strongest associations were recorded at Rio Claro ($\rho = -0.74$ to -0.83) and Santa Gertrudes ($\rho = -0.71$ to -0.75), while Campinas-Centro exhibited the weakest and least consistent relationships ($\rho = -0.15$ to -0.33 , $p > 0.05$ at 250 and

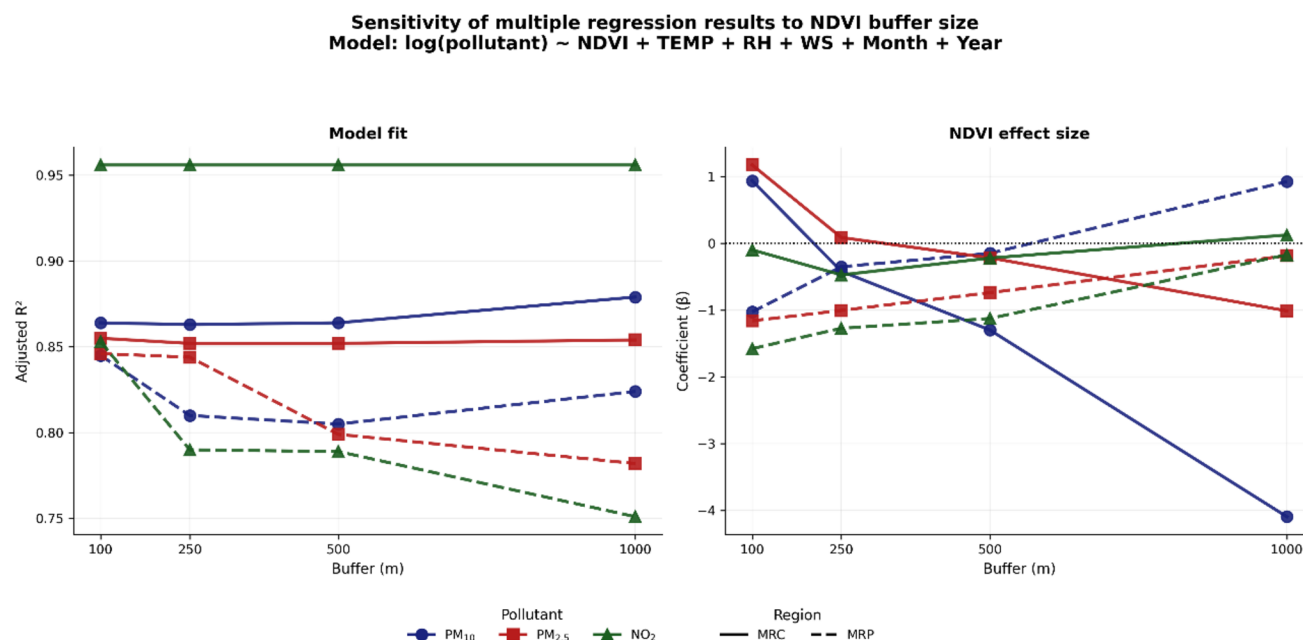


Fig. 6 Sensitivity of multiple regression results to NDVI buffer size for the MLR. The left panel shows adjusted R^2 values across buffer distances (100, 250, 500, and 1000 m) for pollutants at MRC and MRP. The right panel displays the standardized regression coefficients (β) associated with NDVI for each pollutant and region, illustrating how the strength and direction of the NDVI association vary with buffer size

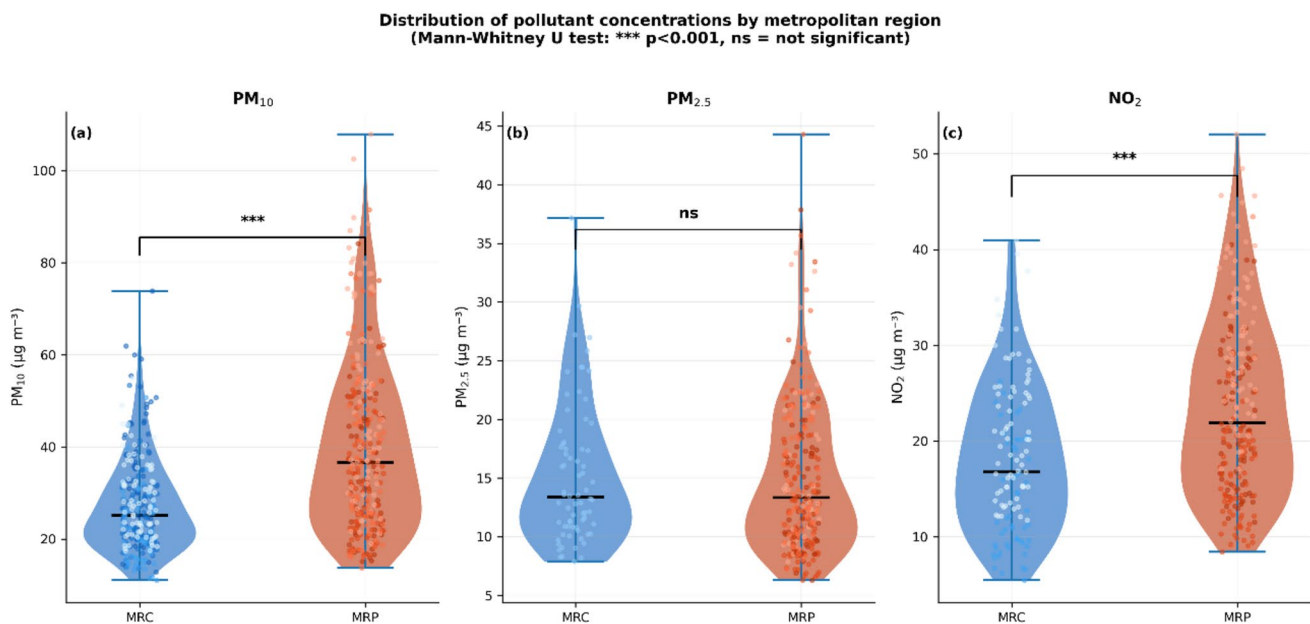


Fig. 7 Distribution of monthly pollutant concentrations by region. Points represent individual monthly observations, and black bars indicate central tendency. Differences between regions were assessed

using the Mann–Whitney U test; *** indicates $p < 0.001$ and ns indicates non-significant differences

500 m buffers). Among MRC stations, Campinas-Taquaral and Americana showed moderate correlations ($\rho = -0.57$ to -0.74 and -0.45 to -0.60 , respectively), whereas MRP stations generally displayed stronger and more spatially stable coefficients.

For $\text{PM}_{2.5}$, the most pronounced correlations were found at Santa Gertrudes ($\rho = -0.72$ to -0.82), Rio Claro ($\rho = -0.58$ to -0.74), and Piracicaba ($\rho = -0.50$ to -0.67), all significant at $p < 0.001$. In contrast, Campinas-V.Uniao (MRC) exhibited weaker associations, particularly at the 250 m buffer ($\rho = -0.17$, $p > 0.05$).

For NO_2 , correlations were moderate, ranging from $\rho = -0.31$ to -0.71 . The strongest associations were recorded at Santa Gertrudes ($\rho = -0.63$ to -0.71) and Piracicaba ($\rho = -0.41$ to -0.43), while Paulínia presented the weakest coefficients ($\rho = -0.31$ to -0.42). MRC stations tended to show slightly attenuated associations compared to their MRP counterparts, potentially reflecting possible differences in urban morphology, traffic density, or the spatial heterogeneity of GI between monitoring regions.

Nevertheless, a tendency toward stronger negative associations at larger buffer zones (500–1000 m) was observed for PM_{10} at several stations, including Rio Claro and Piracicaba.

MLR models were fitted for each pollutant–buffer combination in both regions (Fig. 6). In general, model fit was high, with adjusted R^2 values ranging from 0.75 to 0.96, indicating that the combined predictors explained a substantial proportion of variance in log-transformed pollutant

concentrations. In MRC, PM_{10} and $\text{PM}_{2.5}$ models showed stable performance across buffers (R^2 adj = 0.85–0.88), while NO_2 achieved the highest and most consistent fit (R^2 adj = 0.96), suggesting that meteorological variables and seasonality alone account for most of its variability in this region. In MRP, model performance was slightly lower (R^2 adj = 0.75–0.85) and consistently peaked at the 100 m buffer for all three pollutants (PM_{10} : R^2 adj = 0.84; $\text{PM}_{2.5}$: R^2 adj = 0.85; NO_2 : R^2 adj = 0.85), declining progressively at larger buffer distances.

The adjusted association between NDVI and pollutant concentrations differed markedly between regions. In MRP, NDVI exhibited statistically significant negative coefficients across most buffer scales for all pollutants. Coefficient magnitudes were largest at finer spatial scales, with β coefficients exceeding -1.00 at 100 m for PM_{10} ($\beta = -1.03$), $\text{PM}_{2.5}$ ($\beta = -1.16$), and NO_2 ($\beta = -1.58$), and attenuating progressively with increasing buffer size. For $\text{PM}_{2.5}$ and NO_2 , associations became non-significant at 1000 m ($p = 0.22$ and $p = 0.33$, respectively), indicating that the NDVI–pollutant relationship is most strongly expressed at the local scale. A notable exception was PM_{10} at 1000 m, where a sign reversal was observed ($\beta = +0.92$, $p < 0.001$). In MRC, NDVI coefficients were non-significant across virtually all pollutants and buffer scales ($p = 0.22$ – 0.95), with the single exception of PM_{10} at 1000 m ($\beta = -4.10$, $p = 0.003$). This pattern is likely attributable to the smaller sample size in MRC ($n = 32$ – 70), reduced statistical power, and stronger collinearity

between NDVI and meteorological predictors driven by the shared seasonal cycle.

Pollutant concentration distributions varied between metropolitan regions (Fig. 7). PM_{10} and NO_2 concentrations were significantly higher in the MRP than in the MRC, according to the Mann–Whitney U test ($p < 0.001$). For both pollutants, the MRP exhibited higher central concentrations and a broader distribution, particularly toward higher values. In contrast, $\text{PM}_{2.5}$ concentrations did not differ significantly between regions ($p > 0.05$), suggesting a more similar regional distribution for fine particulate matter compared with PM_{10} and NO_2 .

Discussion

The present study showed that NDVI was predominantly negatively associated with monthly concentrations of PM_{10} , $\text{PM}_{2.5}$, and NO_2 across the two metropolitan regions studied, although the strength and statistical significance of these associations varied among pollutants, buffer sizes, monitoring stations, and regions. These findings support the hypothesis that greener surroundings may be linked to improved local air quality conditions, but also indicate that the NDVI–pollutant relationship is strongly context-dependent. The observed associations were not uniform across space, suggesting that vegetation cover may interact with local emission sources, urban morphology, meteorological conditions, and regional land use patterns.

These findings are consistent with empirical studies reporting inverse associations between urban greenness and air pollutant concentrations in other urban contexts (Dadvand et al. 2015; Matthaïos et al. 2024; Srbínovska et al. 2025). However, direct comparison of effect sizes is limited by differences in pollutants, vegetation metrics, buffer distances, emission sources, and urban form. The buffer-dependent patterns observed here also align with previous evidence that greenness–air pollution relationships are sensitive to spatial scale and exposure definition (Jimenez et al. 2022; Yao et al. 2025). In particular, the stronger adjusted associations at smaller buffers in MRP suggest a more local greenness signal, whereas broader buffer associations, especially for PM_{10} , may also reflect land use gradients rather than vegetation effects alone.

These associations are also compatible with the broader literature on GI and urban air quality, which describes several pathways through which vegetation may influence pollutant concentrations (Nowak et al. 2014; Abhijith et al. 2017). For particulate matter, vegetation surfaces may act as sinks for coarse and fine particles, whereas for NO_2 , uptake can occur partly through stomatal absorption and surface deposition. However, the effectiveness of vegetation depends on canopy structure, leaf traits, species composition, street

geometry, wind flow, and the relative position of sources and receptors (Abhijith et al. 2017). Therefore, NDVI should be interpreted as a broad proxy of vegetation greenness rather than a direct measure of pollutant removal capacity.

The nonlinear pattern observed in the NDVI–pollutant relationships suggests that the association between greenness and air quality may not increase proportionally across the full NDVI gradient. The steep decline in pollutant concentrations at lower NDVI values, followed by a flattening at higher NDVI levels, may indicate diminishing marginal benefits of additional greenness or the influence of other limiting factors, such as pollutant sources, atmospheric stability, and urban form. Similar context-dependent effects have been reported in previous studies showing that vegetation may improve air quality in some settings but may have weak or even adverse effects where dense canopies reduce ventilation in street canyons (Pugh et al. 2012; Setälä et al. 2013; Abhijith et al. 2017). This reinforces the need to consider not only the amount of vegetation but also its spatial configuration and interaction with the built environment.

Seasonality was a major feature of the pollutant time series. Higher PM_{10} , $\text{PM}_{2.5}$, and NO_2 concentrations during the dry season and transition months are consistent with reduced wet deposition, lower atmospheric dispersion, and more frequent biomass burning episodes in southeastern Brazil. In contrast, lower concentrations during the wet season likely reflect enhanced pollutant removal by precipitation and improved atmospheric cleansing. The seasonal anti-phase between NDVI and particulate matter, with lower greenness and higher PM concentrations around the end of the dry season, is environmentally plausible but should be interpreted cautiously. Because both vegetation greenness and pollutant concentrations are strongly seasonal, part of the observed correlation may reflect shared seasonality rather than a direct vegetation effect.

The regional contrast between MRC and MRP further suggests that local LULC and emission profiles modulate the relationship between vegetation and air pollution. PM_{10} and NO_2 concentrations were significantly higher in MRP than in MRC, whereas $\text{PM}_{2.5}$ did not differ significantly between regions. This pattern may indicate that PM_{10} and NO_2 are more strongly influenced by regional or local source structures, including agro-industrial activities, traffic, industrial combustion, soil resuspension, and biomass burning contributions. $\text{PM}_{2.5}$, by contrast, may be more regionally mixed and influenced by secondary aerosol formation or long-range transport, which could explain the weaker regional differentiation. Previous studies in South America have shown that biomass burning emissions can be transported over long distances and substantially affect particulate concentrations far from source regions (Freitas et al. 2005; Artaxo et al. 2013).

The regression models indicated that the adjusted association between NDVI and pollutant concentrations was stronger and more consistent in MRP than in MRC,

especially at smaller buffer distances. This suggests that vegetation surrounding AQMS may be more strongly linked to local air quality variability in MRP, while in MRC the effect of greenness may be partially masked by meteorological controls, urban complexity, or stronger emission heterogeneity. The stronger performance of local-scale buffers in MRP is consistent with the idea that near-station vegetation can influence pollutant concentrations at short spatial scales, particularly where monitoring sites are embedded in less densely urbanized or more heterogeneous landscapes.

However, the sign reversal observed for PM_{10} at the 1000 m buffer indicates that larger-scale NDVI may also capture broader land use contrasts rather than vegetation effects alone. For example, higher NDVI at broader scales may coincide with agricultural areas, unpaved surfaces, biomass burning prone landscapes, or urban–rural gradients, which can contribute to coarse particulate matter. Because traffic density, emission sources, urban morphology, and vegetation structure were not explicitly modeled, these larger-buffer associations should be interpreted cautiously as spatial associations rather than evidence of a specific vegetation-driven mitigation mechanism.

The interpretation of buffer-based NDVI associations is also constrained by the pollutant-specific spatial representativeness of the AQMS. Most station–pollutant combinations included in this study were classified by CETESB as neighborhood-scale, whereas Campinas-Centro was classified as micro-scale for PM_{10} . Therefore, the broader 500 m and 1000 m NDVI buffers should be interpreted primarily as indicators of the surrounding urban-landscape context, rather than as the exact spatial footprint represented by pollutant measurements. This caveat is particularly relevant for micro-scale stations, where concentrations may be strongly influenced by nearby traffic, local emission sources, and site-specific dispersion conditions. Accordingly, broader buffer NDVI associations were interpreted cautiously and not as evidence of direct vegetation-driven pollutant removal.

Several limitations should be acknowledged. First, NDVI should be interpreted as an incomplete proxy for vegetation greenness and not as a direct measure of air quality regulation. It does not capture structural attributes that influence pollutant deposition and dispersion, such as leaf area index, canopy height, species composition, street canyon geometry, aerodynamic roughness, or vegetation arrangement. It also cannot distinguish tree canopy from grasslands, crops, shrubs, or managed green spaces, and may be affected by mixed pixels in dense urban areas.

Second, the analysis was based on AQMS with different pollutant availability and valid sample sizes, which may affect the comparability of results among stations and regions.

Third, multiple station–pollutant–buffer comparisons increase the probability of false-positive results. Therefore,

future analyses should consider false discovery rate correction or hierarchical modeling approaches (Benjamini and Hochberg 1995).

Fourth, because the study period (2019–2025) includes the COVID-19 pandemic and subsequent recovery, emission patterns may have changed due to altered mobility, traffic volumes, and industrial activity. Although the models included month and year fixed effects, meteorological covariates, and HAC standard errors, these adjustments do not explicitly capture intra-year lockdown intensity or municipality-specific recovery dynamics. Therefore, COVID-related non-stationarity is acknowledged as a limitation, and the results should be interpreted as associations over the full study period rather than as stable effects across all pandemic phases.

Finally, although the regression models adjusted for meteorological variables and temporal fixed effects, the observational design does not allow causal inference. Therefore, the results should be interpreted as statistical associations between surrounding greenness and pollutant concentrations, rather than evidence of direct pollutant removal by vegetation.

In general, findings indicate that vegetation cover was often associated with lower pollutant concentrations, but that this relationship depended on pollutant type, spatial scale, regional context, and seasonal dynamics. These results support the relevance of GI as part of air quality management strategies, while also emphasizing that greening policies should be designed with attention to local emission sources, vegetation structure, urban morphology, and meteorological conditions. In rapidly urbanizing and agro-industrial landscapes such as the MRC and MRP, integrating vegetation planning with air quality monitoring and LULC management may provide more effective environmental benefits than increasing greenness alone.

Conclusion

The results indicate that higher NDVI values were generally associated with lower concentrations of PM_{10} , $PM_{2.5}$, and NO_2 , but the strength and consistency of these associations varied substantially by pollutant, region, AQMS, and spatial buffer size.

Seasonality was evident in the observed patterns. Pollutant concentrations increased during the dry season, while NDVI declined, especially near the end of the dry period. This inverse seasonal behavior suggests that part of the NDVI–pollutant association may reflect shared seasonal dynamics rather than a direct vegetation effect. After adjustment for meteorological variables and temporal fixed effects, NDVI showed a more consistent adjusted association with pollutant concentrations in MRP, particularly at smaller buffer distances, whereas in MRC the relationship was weaker and less statistically robust.

The findings reinforce the relevance of spatially explicit vegetation indicators for urban and regional air quality studies, but they also show that greenness alone cannot fully explain pollutant variability. Local emission sources, land use composition, agricultural activities, urban morphology, atmospheric dispersion, and vegetation type likely mediate the observed relationships. Therefore, these findings support considering GI as one component of integrated air quality management, rather than as an isolated mitigation strategy.

Future research should incorporate more detailed land cover classifications, vegetation structure metrics, traffic and emission-source indicators, and additional meteorological variables such as wind direction. Statistical approaches that explicitly account for repeated observations, station-level heterogeneity, autocorrelation, and multiple testing would further strengthen future analyses. In general, this study provides evidence that vegetation greenness is associated with air quality patterns in two contrasting metropolitan regions, while highlighting the need for cautious interpretation and context-specific planning.

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Data availability The materials for this study will be made available upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

Ethical approval This study did not involve human participants, animals, or identifiable personal data. Therefore, ethics committee approval was not required.

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